Live wire detector

Find 'em concealed wires without electrocuting yourself

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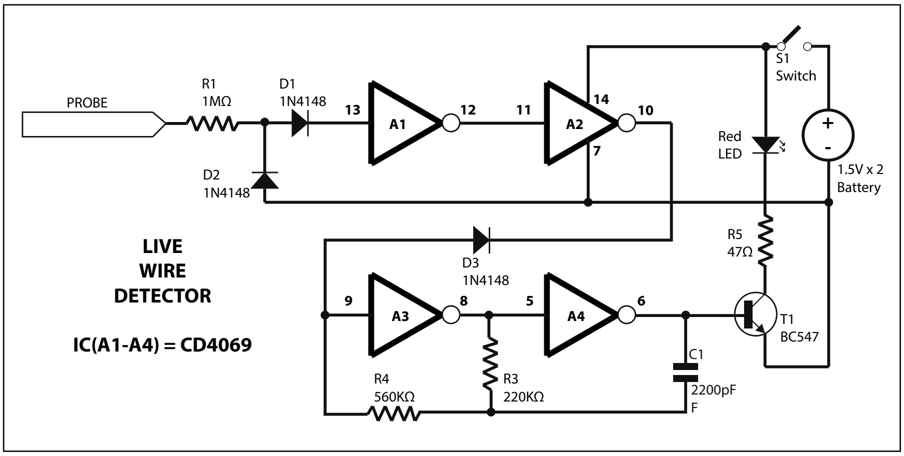
There are instances when you wish you were like Raiden (or Pikachu for those born in the late 90's) and have the ability to sense and manipulate electricity. We have something that can sense and let you know if there is a live wire concealed underneath something(wall/plaster/floor). This nifty little gadget is way helpful if you've recently moved to a new place and are wary as to where all the power lines exist in the house. A more dangerous use is to detect water pipes by passing current through the pipes, however, electricity takes the path of least resistance, thus only one route will be noticed. **Do not attempt this at all as this is very dangerous and you can electrocute anyone who uses the same water source,** especially if you happen to live in an apartment complex. Stick to a metal detector if you wish to detect concealed water pipes.

These nifty things come in various avatars under various names, common being EMF detector (technically most accurate), broken wire detector etc. Powerful and much more sensitive variants of the circuit are used by construction staff. The Bosch DMD4K is one such tool, however, forthe purpose of this article we'll concentrate on just the EMF detection.

The working is pretty simple, a probe is made using a strand of copper wire which is brought near the live power source. The first two gates A1 and A2 buffers the voltage picked up by the probe. In order to increase sensitivity of the probe you can coil the copper strand.

What you'll need

Component	Value
R1, R3, R4, R5	1M , 220K, 560K , 47
IC (A1-A4)	CD4069
D1, D2, D3	1N4148 diodes
C1	2200pF capacitor
T1	BC547 NPN Transistor
2 Batteries	AA / AG13
LED	Any colour
S1	Switch (Push to ON)



Circuit diagram

Gates A3 and A4 generate a constant pulse with the help of resistors R3, R4 and capacitor C1.

Whenever any AC voltage is picked up, during the positive half of the cycle, the output at pin no 10 of the IC goes high, which turns on the oscillator part of the circuit (i.e. A3 and A4), this makes the LED blink at whatever frequency the oscillator is running, in this case 1000 Hz. So the LED seems to stay on to the human eye when actually it's blinking rapidly.

Never touch the probe of the circuit to an exposed live wire under any circumstances. The sensitivity of the circuit depends on the probe at the end. If you happen to find a phone's pick up coil then you can significantly increase the sensitivity. You can come with various ways of implementing this circuit, if you wish you can add a decade counter and modify the circuit so that the closer you get to the live wire, the more LEDs you can have turning on simultaneously. Then you can combine four such circuits placed at four corners of a square to create a tool which can tell you the alignment of the concealed live wire. Here are a few tools which have implemented the same. http://dgit.in/PYShOO http://dgit.in/UlLNaX.

Another great circuit that we came across was using Arduino here at http://dgit.in/RB1SNU which utilises an LED whose brightness is varied according to the distance from the live wire. An improved version of the same is also featured on Make magazine's Youtube channel - http://dgit.in/Tai6P7 This version uses a LED bar graph which shows the level of the EMF in the vicinity, it's not accurate andthus, isn't indicative of any actual values, This is similar to using a decade counter but way easier and better looking. Experiment and modify the design to come up with more useful gadgets.